University of Cape Town
Department of Computer Science

Computer Science CSC3003S
November Examination, 2021

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of **Four Sections:**

Section A: *Advanced Software Design 1*

Section B: *Advanced Software Design 2*

Section C: *Theory of Algorithms*

Section D: *Theory of Computation*

Total Marks: 90	Maximum marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in FOUR separate books. Use one book for Advanced Software Design 1, one book for Advanced Software Design 2, and one book each for Theory of Algorithms and Theory of Computation. **Label each of the four books as ASD1, ASD2, ToA and ToC**.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular, "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book.

Please **do this for the front cover of the different books used for each section.**

This is needed so that we can allocate marks and not miss questions. **A list of sub-questions is *not* required.**

NOTE: The "Master Theorem" can be found at the end of the paper.

Section A. [22]

Question 1. **[22]**

Refer to the excerpt below and answer the questions relating to the software development methods:

You have just completed a meeting with a prospective client. This client has asked you to develop a system that enables employees to log in with a username and password which must include two factor authentication (2FA). The system must allow employees to log their hours worked by allowing them to select their employee number and then input their hours worked. The system will then compute their weekly pay at the end of every week. The customer has informed you that you have exactly 6 months to complete the project.

a) Sketch out a high-level project plan using a waterfall approach. [2]

b) Sketch out a high-level project plan using Rational Unified Process. [2]

c) What project management approach would you recommend to the customer for this client? Give at least three (3) reasons for your choice, clearly comparing this approach to alternatives. [5]

d) Your customer has since asked you to develop a prototype that demonstrates exactly how only the login process would function – describe the type of prototype you are likely to use and motivate your decision by explaining why other types of prototypes will not be suitable. [3]

e) What is a burndown chart and how is it used? [2]

f) You have been asked to test the functionality of the username and password authentication method. What type of test would you recommend? Motivate your answer. [3]

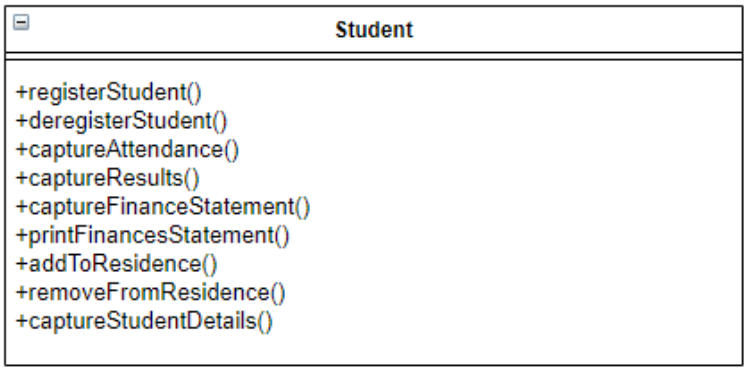
g) You have decided to take an agile approach to the project, and you have weekly meetings with your client regarding the look and feel of the login process. Which principle of the Agile manifesto does this reflect? Explain your answer. [2]

h) You are considering using open source libraries to support your implementation of the system. What are some of the issues (at least three) you need to consider and address before you finalise your decision? [3]

Section B. [23]

Question 2. [5]

You are building a university information system to track student registration, finances, results, attendance, and residence occupancy. Refer to the class diagram below and answer the questions that follow.



Refactor this class to adhere to the **SINGLE RESPONSIBILTY PRINCIPLE (SRP)** by drawing a revised class diagram, showing both the classes and corresponding methods. Explain why you structured your classes in the manner you have and specify the responsibility of each class in your diagram.

Question 3. [7]

You have been contracted to design an ecommerce system that needs to sell different categories of products to clients. The products will all have a picture, description, unit price and rating section, regardless of the category. There are thousands of products, but your system has limited memory; creation of the product objects should be done on the fly. Consider how you will implement this system using design patterns.

a) What is a design pattern? [3]

b) Which design pattern would you recommend? Explain how the pattern works, and motivate your choice based on the scenario. [4]

Question 4. [7]

You are the architect contracted to create a system to allow a university to analyse their students' results. This system will make use of the same large volume of shared data to provide several different views of the data. For example, the system should cater for showing the basic descriptive statistics such as the mean, min and max marks for a particular assessment; however the teaching and learning team would also like to see how many students attempted an assessment versus how many students passed. There are also future plans to expand this system.

a) Sketch an architectural diagram for the system and justify it. [5]

b) Highlight any potential pitfalls you will consider while implementing this solution. [2]

Question 5. **[4]**

Your company has recently adopted an Agile methodology and is considering implementing DevOps. The CIO has asked you for your recommendations on the matter. Would you recommend DevOps for an Agile environment? Motivate your answer.

Section C. [22]

Question 6. **[6]**

Answer the following multiple choice questions (2 marks each):

a) Given the following recurrence relation for the number of operations of a recursive algorithm:

$M(n) = 3\ M(n/3) + n$
$M(1) = 0$

What is its big-Θ efficiency?

a. $\Theta(n)$
b. $\Theta(n \log n)$
c. $\Theta(n^2)$
d. $\Theta(n^2 \log n)$
e. $\Theta(n^{\log 3})$

b) What is the efficiency class of the following algorithm?

```
val = 1
FOR i = 1 TO n
   FOR j = 1 TO i
      val = val * i * j
FOR k = n DOWNTO 1
   val = val + k
RETURN val
```

a. $O(1)$
b. $O(\log n)$
c. $O(n \log n)$
d. $O(n^2)$
e. $O(n^3)$

c) What problem does the following algorithm solve?

```
Mystery( x[0..n-1], y[0..n-1])
   m = inf
   FOR i = 0 TO n-1 DO
      FOR j = i+1 to n-1 DO
         d = sqrt(pow(x[i]-x[j], 2) + pow(y[i]-y[j], 2))
         IF d < m THEN
            m = d
            i1 = i
            i2 = j
   RETURN i1, i2
```

a. Transitive Closure
b. Closest Pair
c. Knapsack Problem
d. All Pairs Shortest Path
e. String Matching

Question 7. **[4]**

Answer the following multiple choice questions (1 mark each):

a) If an algorithm has a running time as n (the number of inputs) gets larger that is *exactly* proportional to g(log n) then its computational complexity is:

a. O(log n)
b. O(n)
c. Θ(log n)
d. Θ(n log n)
e. Ω(n log n)

b) What well known problem best fits the following description?

Given a set of items each with a weight and a value, find the subset of those items with the highest total value whose total weight falls below a specified maximum.

a. Convex Hull
b. The Travelling Salesman Problem
c. Transitive Closure
d. String Matching
e. The Knapsack Problem

c) In what class of algorithms is Sorting by Counting?

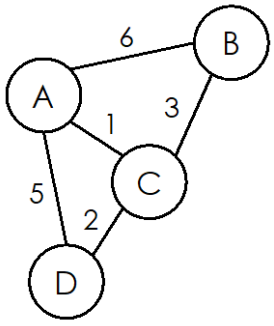
a. Brute Force
b. Divide and Conquer
c. Space-Time Tradeoffs
d. Dynamic Programming
e. Greedy Technique

d) For inputs of size n and in the worst case, which of these algorithms has the best big-O time efficiency?

a. Binary Search of a sorted list
b. Warshall's Algorithm
c. Closest Pairs by Brute Force
d. Horner's Rule
e. Quicksort

Question 8. **[6]**

Show the steps of Floyd's Algorithm for finding the shortest path between every pair of vertices as applied to the following undirected graph:



Question 9. **[6]**

You are given a street with buildings along it and are asked to determine the height of the building on the left-hand side for any given position along the street. Assume that the start of the street is represented as 0 and the end as d (measured in km). Each building is represented by 3 values: (start, end, height) representing the position along the street where the building begins, where it ends, and its height, respectively. (We assume that the height of the building does not vary along its façade.)

Note that this height query will be run multiple times for a given street.

a) Outline a brute force algorithm (in English or pseudocode) to solve this problem. [2]

b) Now give pseudocode for a more efficient algorithm to solve this problem. [3]

c) Give the efficiency of each of the two algorithms above. [1]

Section D. [23]

Question 10. **[2]**

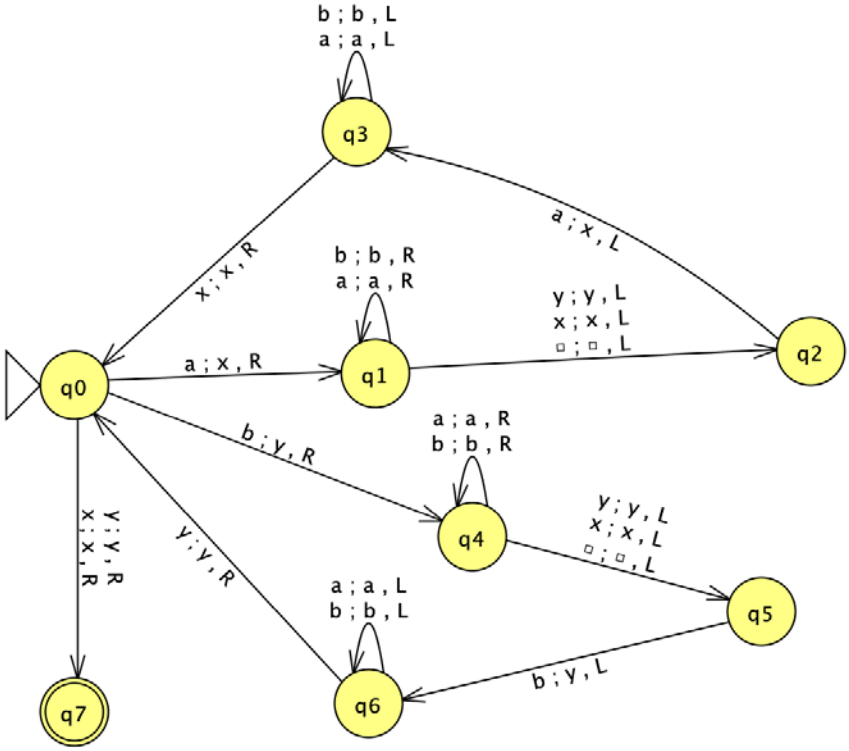
Consider the following transition for some Turing machine.

$\delta(p, [x,0]) = \{(q, [B, 0], R), (r, [B,1], L)\}$

Describe in your own words what sort of Turing machine it is and what it does.

Question 11. **[2.5]**

Describe the following Turing machine in one sentence, either in terms of the language it accepts or in terms of the problem it solves.



Question 12. **[7]**

One can design a Turing Machine that satisfies the specification “copy the input to its adjacent blanks on the right-hand-side on the tape” as a straightforward plain deterministic Turing Machine, a multiple-tracks Turing Machine with storage, or a multiple-tapes Turing Machine. Discuss how each option affects the running time. Note: you do not need to design each machine in detail with all the transitions; just the general solution approach for solving the task for each type of machine suffices for providing evidence.

Question 13. **[4]**

Assume there is a program halts(P,X) that takes as arguments a program P and an input X to P, and determines whether P will halt on X or not, returning true if P halts on X, and false otherwise; this assumption that halts(P,X) exists may, or may not, be correct.
Consider another program, LOOP(X), that takes as input a program X and calls halts(X,X):

```
A: if halts(X,X) then goto A; else halt.
```

halts(X,X) will determine whether X will halt when presented with itself as input. If X halts on input X, then LOOP(X) will loop for ever, otherwise LOOP(X) will halt.

a) Does LOOP(LOOP) halt? [1]

b) What does that say about our assumption about program halts(P,X)? [3]

Question 14. **[7.5]**

There are six languages (or equivalently, problems) A, B, C, D, E, and F. All we know about them is the following:

- A is in P.
- B is in NP.
- C is NP-complete.
- D is Recursive.
- E is Recursively Enumerable but not Recursive.
- F is non-Recursively Enumerable.

For each of the five statements below, tell whether it is:

- CERTAIN to be true, regardless of what problems A through F are and regardless of the resolution of unknown relationships among complexity classes (such as “is P = NP?”).
- MAYBE true, depending on what languages A through F are, and/or depending on the resolution of unknown relationships such as P = NP?
- NEVER true, regardless of what A through F are and regardless of the resolution of unknown relationships such as P = NP?

and, for questions a to e, *briefly* explain why for each case.

a) There is a reduction from E to D. [1.5]

b) There is a polytime reduction from C to B. [1.5]

c) There is a polytime reduction from B to $\bar{C}$. [1.5]

d) There is a reduction from D to C. [1.5]

e) There is a reduction from F to A. [1.5]

Appendix A – Master Theorem

If we have a recurrence of this form: $T(n) = aT(n/b) + f(n)$, $f(n) \in \Theta(n^d)$ and $T(1) = c$ then

$T(n) \in \Theta(n^d)$ if $a < b^d$

$T(n) \in \Theta(n^d \log n)$ if $a = b^d$

$T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$

-END-